

Is the Sun a Clock?

Renewable Composition and the Value of Information in Electricity Storage Arbitrage

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Abstract

Grid-scale storage in Europe is financed on the expectation that renewables will create arbitrage value. We ask a prior question: whether that value can be captured without forecasting, and whether the answer depends on which renewable creates it. Using hourly day-ahead prices for 19 European bidding zones from 2019 to 2026, we operate a price-taking battery four ways, differing only in information: perfect foresight, a typical-day schedule learned from the previous four weeks, persistence, and a forecaster built on TSO day-ahead fundamentals. Identification comes from daily forecast wind and solar penetration under zone-by-year, zone-by-month and date fixed effects, so the treatments vary with weather alone. Solar creates arbitrage value: 10 percentage points more forecast solar raises a day's perfect-foresight value by 28%, while wind raises it not at all. Solar makes the day's price shape more typical and wind makes it less so, by equal and opposite amounts per standard deviation of weather. Calendar-based operators capture a larger share of solar-driven value, by about four points per 10 pp, an effect with stable magnitude but marginal pre-registered significance ($p = 0.10$). Wind's cost to them grows with a system's wind share. Storage value created by solar is largely public; value created by wind increasingly rewards forecasting. All hypotheses were pre-registered.

1 Introduction

Grid-scale batteries are being built across Europe on the expectation that the price volatility created by wind and solar will pay for them. That expectation has two parts that are usually run together. The first is that renewables create arbitrage value: they depress prices when they produce and leave scarcity when they do not, widening the intraday spreads a battery trades on. The second, rarely stated, is that the value can be captured by whoever owns the battery. This paper is about the second part. Arbitrage value exists only in hindsight. An operator commits its schedule in the day-ahead auction before prices are known. What it earns depends on how much of the day's price shape it can anticipate.

The information needed to anticipate that shape depends, we argue, on *which* renewable creates it. Solar output follows the sun. On a clear day it carves a trough into the middle of the day at the same clock hours as on every other clear day, so the value it creates recurs at predictable times: a schedule learned from recent weeks will largely find it. Wind output follows weather systems that arrive and depart at no particular hour, so the value it creates sits wherever the weather puts it, and capturing it requires knowing the day. If this is right, the two technologies of decarbonisation differ in more than their levelised cost and value factor. Solar creates storage value that is *public*,

available to any operator who knows the calendar. Wind creates storage value that is *informational*, available only to operators who forecast.

That distinction matters beyond the private economics of batteries. If the value of storage increasingly depends on forecasting capability, the returns to operating storage accrue to firms with scale in data, modelling and trading. That has implications for the market structure of flexibility, and for the public value of the forecasts that transmission system operators already publish. It also implies that a system decarbonising mainly through wind and one decarbonising mainly through solar will reward different kinds of storage operator.

We test the argument on hourly day-ahead prices for 19 European bidding zones from 2019 to 2026, a panel spanning Danish and German wind, Iberian and Greek solar, French nuclear and Polish coal. We model a price-taking 1 MW, 2-hour battery operated four ways, which differ only in the information used to set the schedule:

1. **Perfect foresight**, which defines the ceiling.
2. **Typical day**, which optimises against the mean price profile of the previous four weeks, and so knows the calendar and nothing about the day.
3. **Persistence**, which optimises against yesterday's prices.
4. **Forecaster**, which adds a gradient-boosted correction built from the TSOs' day-ahead wind, solar and load forecasts.

The share of the ceiling each operator captures measures how much of the day's value its information set reaches. Our treatments are the TSOs' own day-ahead forecasts of wind and solar output, which are the information on which the auction clears. Identification comes from weather. With zone-by-year, zone-by-month and date fixed effects, the remaining variation in forecast penetration is day-to-day weather within a zone's year and season, relative to other zones on the same day. This design is a response to the central difficulty of the question. Renewable penetration trends upward everywhere, alongside everything else that changed over the decade, and a specification that lets that trend through will find an association between any two series that happen to rise together.

We find that the two technologies differ sharply, though not entirely as hypothesised.

- **Solar creates arbitrage value; wind does not.** A day with 10 percentage points more forecast solar offers 28% more perfect-foresight value and a €22/MWh wider spread. A windier day offers no more value, because wind lowers the whole price curve more than it reshapes it.
- **Solar makes the day's price shape more typical, and wind makes it less typical.** Per standard deviation of weather the two effects are equal and opposite. This is the mechanism the hypothesis proposed, and it is precisely estimated.
- **Calendar-based operators capture a larger share of solar-driven value.** The wind-minus-solar difference is about four points of capture per 10 pp, and its magnitude is stable across every robustness check and every leave-one-zone-out fit. At the pre-registered specification it is significant only at the 10% level. The difference comes mainly from solar *raising* calendar capture rather than wind lowering it.
- **Wind's cost to calendar operators grows with wind's share of the system.** It is negligible at

the panel average and appears clearly in wind-dominated zones like Denmark.

The sun, in short, behaves like a clock. Whether the wind behaves like a lottery depends on how much of it a system has.

The paper makes three contributions:

1. **Existing work computes the perfect-foresight value of storage arbitrage across European zones** (Mercier, Olivier, and De Jaeger 2023), or measures forecast-based capture within one or two markets (Hornek et al. 2025; Falezza 2026). We combine an operator ladder with cross-market variation to ask where the value of information *comes from*.
2. **We show that the renewable technology matters**, extending to storage the literature showing that wind and solar affect price formation asymmetrically (Hirth 2013; López Prol, Steininger, and Zilberman 2020).
3. **We provide a design for identifying renewable effects from weather** that is robust to the common trends that have undermined time-series evidence on this question. All hypotheses and specifications were registered before any outcome was computed.

Section 2 reviews related work. Section 3 describes the data and the operators, and Section 4 the empirical strategy. Section 5 presents results, Section 6 robustness and limitations, and Section 7 concludes.

2 Related literature

This paper connects three literatures that have developed largely separately: the valuation of storage arbitrage, the effect of variable renewables on wholesale price formation, and electricity price forecasting.

Storage arbitrage valuation. The standard approach values a price-taking store by optimising its dispatch against historical prices under perfect foresight. Sioshansi et al. (2009) apply it to PJM and also estimate the welfare effects of storage at scale; Walawalkar, Apt, and Mancini (2007) value energy arbitrage and regulation services in New York; and Sioshansi (2010) examine how ownership structure changes the welfare calculus. For Europe, Zafeirakis et al. (2016) backtest arbitrage strategies across day-ahead markets. They find value concentrated in less integrated, import-dependent and less competitive markets, which attributes the cross-market dispersion to market structure. Closest to our setting, Mercier, Olivier, and De Jaeger (2023) compute the perfect-foresight value of a price-taking battery for every EU-28 bidding zone, plus Norway, Switzerland and Turkey, varying efficiency and duration. Our ceiling V^* is their object. We take it as the denominator and ask what share of it operators with less information realise.

A smaller and more recent literature measures that share directly, within a single market. Hornek et al. (2025) find that a forecast-driven battery captures around 89% of perfect-foresight revenue in the German continuous intraday market. Falezza (2026), trading German and Swiss day-ahead, intraday and reserve markets, finds that capture depends on the rank correlation between forecast and realised prices rather than on conventional error metrics. Forecasts above a high rank-

correlation threshold capture almost all of the perfect-foresight revenue, while persistence captures about a third. This is the single-market counterpart to our mechanism. Our shape-novelty variable measures, day by day, how far the realised price shape departs from the shape a calendar-based operator expects. What neither paper can do with one or two markets is ask how capture varies with the generation mix, which is the variation our panel supplies.

Renewables and price formation. Variable renewables enter the merit order at near-zero marginal cost and depress prices when they produce (Sensfuß, Ragwitz, and Genoese 2008). Regression evidence for Germany puts the effect at roughly €1 per MWh for each additional GWh of wind and solar (Cludius et al. 2014). Ketterer (2014) shows that German wind lowers the price level but raises its volatility, and volatility is the raw material of arbitrage. Because renewables depress prices precisely when they produce, their own market value falls with penetration (Hirth 2013). In California, wind and solar cannibalise each other asymmetrically: wind penetration lowers solar’s value factor while solar penetration raises wind’s (López Prol, Steininger, and Zilberman 2020). That asymmetry between the two technologies in price formation is the price-side analogue of the asymmetry we test on the storage side. The merit-order literature is concerned mainly with the *level* of prices. Our outcome depends on their *intraday shape*, and specifically on whether that shape is predictable from the calendar.

Renewables and the value of flexibility. If renewables raise volatility, they should raise the value of storage, and equilibrium studies confirm that storage investment responds to renewable-driven volatility. Butters, Dorsey, and Gowrisankaran (2025) model battery entry in California, finding it economic without subsidy only at high renewable shares, with the first tranches of storage doing most to reduce prices. Karaduman (2023) finds a non-monotonic relationship in South Australia: storage first erodes renewable revenues by flattening prices, then raises them once renewable capacity is large enough to face curtailment. Hydro reservoirs have long performed this role. Danish exports track Nordic reservoir levels rather than wind directly (Green and Vasilakos 2012), and wind shifts the shadow value of stored water in Norwegian hydro areas (Mauritzen 2013). These studies establish that renewables change *how much* flexibility is worth. Our question is *who can capture it*: whether the value renewables create is available to any operator who knows the season, or only to one who knows the day.

Price forecasting. Electricity price forecasting is a mature field (Weron 2014). Recent work stresses rigorous, comparable benchmarking (Lago et al. 2021), and shows that European forecasts improve when neighbouring coupled markets are included (Lago et al. 2018). That last finding matters for identification here. Under market coupling, a zone’s price responds to its neighbours’ weather, so our wind coefficient measures the effect of wind weather in a zone’s region rather than of strictly domestic output. We return to this in Section 6. We use forecasting instrumentally. Our forecaster exists to measure how much of the value lost by a calendar-based operator is recoverable with information available at gate closure, not to compete on forecast accuracy.

Inference. Our design has a moderate number of natural clusters: 18 bidding zones in the estimation sample. Cluster-robust inference is unreliable at this size (Cameron, Gelbach, and Miller 2008), and the wild cluster bootstrap with six-point weights gives more reliable p-values (Webb

2023). We follow that recommendation. Our treatments are continuous, not staggered binary adoptions, so the negative-weighting problems of two-way fixed effects under staggered timing (de Chaisemartin and D’Haultfoeuille 2020; Goodman-Bacon 2021) do not arise in their canonical form. Heterogeneity in effects across zones remains a concern, which we address with leave-one-zone-out estimates.

Data. The ENTSO-E Transparency Platform is the only harmonised public source for this panel, and it has documented quality problems: inconsistent labelling, gaps and revisions (Hirth, Mühlenpfordt, and Bulkeley 2018). We report our coverage audit and exclusions in full in Section 3.

3 Data

3.1 Market data

All market data come from the ENTSO-E Transparency Platform, which publishes standardised data for European bidding zones under Commission Regulation (EU) No 543/2013. We use five series at hourly resolution for 2019-01-01 to 2026-09-20:

- day-ahead clearing prices;
- actual total load;
- actual generation by production type;
- the transmission system operators’ (TSOs’) day-ahead load forecasts;
- the TSOs’ day-ahead forecasts of wind and solar generation.

The unit of observation is the **bidding zone**, not the country, because day-ahead prices clear at zone level. Denmark’s two zones sit on different synchronous areas and frequently clear at different prices.

The sample comprises 19 zones chosen to span the range of European generation mixes:

Type	Zones
Wind-heavy	DK_1, DK_2, DE_LU, NL, BE, PT, ES
Solar-growing	GR, IT_NORD
Nuclear-dominated	FR
Coal-heavy	PL, CZ
Hydro and nuclear	AT, CH, NO_2, SE_3, SE_4, FI, EE

Great Britain is excluded because its ENTSO-E coverage ends in 2020–21. Ireland (SEM) is excluded because its load forecast is almost entirely absent after mid-2021.

Two features of the data require care:

- **Resolution.** From October 2025 the day-ahead auction moved to a 15-minute market time unit in most zones. We average quarter-hours into hours, which is the price a battery holding a flat position over the hour receives.

- **The trading day.** Every coupled zone trades a *delivery day* defined on Central European Time, including zones whose civil time differs. In summer, Portugal and Ireland begin their trading day at 23:00 local time and Finland at 01:00. We verified this against the platform’s own price documents. Every “day” in this paper is a CET delivery day: 24 hours, or 23 or 25 on clock-change days.

3.2 Treatment variables

Our treatments are the TSOs’ day-ahead forecasts, not realised generation. The day-ahead price clears at 12:00 CET on $D-1$, before realised output is known, so forecasts are the information on which the price is formed. For zone z and delivery day d :

$$\text{wind}_{zd} = 100 \times \frac{\sum_{h \in d} \widehat{W}_{zh}}{\sum_{h \in d} \widehat{L}_{zh}}, \quad \text{solar}_{zd} = 100 \times \frac{\sum_{h \in d} \widehat{S}_{zh}}{\sum_{h \in d} \widehat{L}_{zh}},$$

where $\widehat{W}$, $\widehat{S}$ and $\widehat{L}$ are the forecast wind output, solar output and load in MWh. A penetration is defined only when the forecasts cover every hour of the day. Czechia publishes no wind forecast, so it drops out of the estimation sample, which has 18 zones. It remains in the descriptive statistics.

3.3 Arbitrage operators

We model a price-taking battery with 1 MW of power, 2 MWh of energy and 85% round-trip efficiency. Batteries of one and four hours are robustness checks. Each day it starts and ends empty, which makes days independent and comparable. For any price vector $\pi = (\pi_1, \dots, \pi_H)$ over the H hours of a day, the schedule solves the mixed-integer linear programme

$$\max_{c,s,e,z} \sum_{h=1}^H \pi_h (s_h - c_h) \quad \text{s.t.} \quad e_h = e_{h-1} + \sqrt{\rho} c_h - s_h / \sqrt{\rho}, \quad 0 \leq e_h \leq \bar{E}, \quad 0 \leq c_h \leq \bar{P} z_h, \quad 0 \leq s_h \leq \bar{P} (1 - z_h), \quad e_0 = e_H = 0,$$

with c charging and s discharging power, e the stored energy, ρ the round-trip efficiency, and binary z_h preventing simultaneous charging and discharging. The binaries are necessary rather than cosmetic. At negative prices, charging and discharging at the same time is strictly profitable in the linear relaxation, because it burns energy the operator is paid to take.

The four operators differ **only** in the price vector they optimise against. Each schedule is then settled at the day’s actual prices.

Operator	Optimises against	Information used
Perfect foresight (PF)	Actual prices of day d	The ceiling, V_{zd}^*
Typical day (TD)	Mean price by market hour over days $d - 28$ to $d - 1$	The recent calendar shape only

Operator	Optimises against	Information used
Persistence (PS)	Day $d - 1$'s prices, by market hour	Yesterday only

For TD, at least 20 of the 28 days must be complete. Every input is known by gate closure: day $d - 1$ cleared on $d - 2$. Because every operator shares the same optimiser, the gap between any two is a pure difference in information. The **capture share** $100 \times V_{zd}^k / V_{zd}^*$ measures how much of the available value operator k realises.

The TD operator is the conceptual centre. It knows everything about a day that its date reveals: the season, the weekday pattern and the recent diurnal shape, including the solar trough if the sun has been shining. It knows nothing specific to the day. Its shortfall from the ceiling is therefore the part of arbitrage value that requires day-specific information.

3.4 Outcomes

- **Capture shares** of TD and PS. Days whose ceiling falls below the zone's 5th percentile are excluded, because a ratio with a near-zero denominator is meaningless.
- **Scale:** $\log V^*$.
- **Mechanism.**
 - *Shape novelty:* $1 - \text{corr}(\pi_d, \bar{\pi}_d^{TD})$, the extent to which the day's price shape departs from the typical day's.
 - The *daily spread*, $\max_h \pi_h - \min_h \pi_h$.

4 Empirical strategy

We estimate

$$Y_{zd} = \beta_w \text{wind}_{zd} + \beta_s \text{solar}_{zd} + \alpha_{z,y(d)} + \gamma_{z,m(d)} + \lambda_d + \varepsilon_{zd}. \quad (1)$$

The three sets of fixed effects each remove a named threat:

- **Zone-by-year effects** $\alpha_{z,y}$ absorb capacity build-out, market reforms and any zone-specific trend. Installed capacity is endogenous, and a penetration that trends alongside everything else in the sample is exactly the variation that produced spurious associations in earlier single-market specifications of this question (Appendix A).
- **Zone-by-month effects** $\gamma_{z,m}$ absorb each zone's seasonal cycle, including the solar season and the hydrological year.
- **Date effects** λ_d absorb everything common to Europe on a given day: gas and carbon prices, the 2021–23 energy crisis, and the continental component of weather.

The remaining variation in wind_{zd} and solar_{zd} is day-to-day weather within a zone-year-month, relative to other zones on the same day. Weather is not chosen by any market participant, so we interpret β as the causal effect of a day’s forecast renewable output on the outcome.

The **key hypothesis** (H1) is $\beta_w < \beta_s$ for TD capture: wind lowers the share of value capturable from the calendar more than solar does. To test it as a single coefficient, we re-parameterise (1) with wind_{zd} and total variable renewable penetration $\text{vre}_{zd} = \text{wind}_{zd} + \text{solar}_{zd}$ as regressors. The coefficient on wind is then exactly $\beta_w - \beta_s$.

Inference. Errors are clustered by zone. With 18 clusters, conventional cluster-robust inference over-rejects (Cameron, Gelbach, and Miller 2008; MacKinnon and Webb 2018). We therefore report wild cluster restricted bootstrap p-values with Webb six-point weights (Webb 2023) and 9,999 replications, computed on the fixed-effect-demeaned data.

Placebo. Adding the forecast error in wind (realised minus forecast, relative to forecast load) should leave every day-ahead outcome unchanged, because the auction clears before the error is realised. A non-zero coefficient would indicate a timing error in the data, or a forecast series that embeds information unavailable at gate closure.

All hypotheses, specifications and robustness checks were registered before any panel outcome was computed (commit 6aa7cdc). Deviations are reported in Appendix A.

5 Results

5.1 How much value there is, and who captures it

Table 1 summarises the panel. A 1 MW, 2-hour battery with perfect foresight of day-ahead prices would have earned €153 per day on average across the 19 zones, or about €56,000 per MW per year from day-ahead arbitrage alone. This is a ceiling, not a forecast of revenue. The ceiling varies more than threefold across zones:

- **Lowest:** Norway’s reservoir-hydro zone NO_2 (€68), where hydro already smooths prices and 22% of days have perfectly flat prices and zero value.
- **Highest:** Estonia (€240) and Greece (€208).

Table 1: Descriptive statistics by bidding zone, sorted by mean forecast wind penetration.

Zone	Wind (%)	Solar (%)	V* (€/MW/day)	TD (%)	PS (%)	FC (%)	Novelty	Days
DK_1	58.4	8.8	162	77.0	64.8	78.3	0.247	2,814
DK_2	29.0	6.3	175	76.8	67.4	78.0	0.246	2,814
DE_LU	26.9	12.9	180	83.1	74.7	88.9	0.197	2,814
PT	25.5	7.0	107	74.7	66.9	78.8	0.198	2,820
ES	24.2	15.6	112	76.1	69.7	83.2	0.191	2,814
SE_4	22.1	3.6	169	75.3	65.2	74.9	0.260	2,814
GR	19.9	16.8	208	69.1	60.6	74.1	0.247	2,820

Zone	Wind (%)	Solar (%)	V* (€/MW/day)	TD (%)	PS (%)	FC (%)	Novelty	Days
FI	16.1	1.1	148	67.5	53.7	66.3	0.298	2,820
NL	15.9	8.6	190	80.0	70.2	83.1	0.193	2,814
BE	15.0	9.0	175	80.6	70.3	82.4	0.193	2,814
AT	13.6	6.0	156	80.8	71.8	82.9	0.181	2,814
PL	11.5	8.7	158	71.2	59.2	74.7	0.188	2,814
NO_2	11.2	0.0	68	17.5	11.3	19.4	0.312	2,814
SE_3	10.7	1.6	130	72.4	61.9	71.4	0.261	2,814
EE	10.0	7.4	240	69.9	57.0	69.3	0.253	2,820
FR	9.7	5.3	151	81.4	72.6	83.6	0.188	2,814
CH	0.1	4.0	100	77.7	70.5	81.7	0.167	2,813
IT_NORD	0.1	5.9	106	75.9	69.3	82.4	0.177	2,814
CZ	nan	5.4	174	80.3	71.7	83.4	0.181	2,814

Notes: Means over 2019-01-01 to 2026-09-20. Wind and solar are TSO day-ahead forecast output as a share of forecast load. V* is the perfect-foresight arbitrage value of a 1 MW / 2 MWh battery at 85% round-trip efficiency. TD, PS and FC are capture shares of V* for the typical-day, persistence and forecaster operators, on days above each zone's 5th percentile of V*. Novelty is $1 - \text{corr}(\text{day's prices}, \text{typical-day profile})$. CZ publishes no wind forecast and is outside the estimation sample.

Figure 1 illustrates the mechanism the paper tests. It shows two days in Germany-Luxembourg in 2024, selected mechanically as the days with the highest forecast wind and the highest forecast solar penetration. On the windiest day (24 November, wind forecast at 93% of load), prices sat near zero for the whole day. The typical evening peak simply did not happen, and there was almost nothing to arbitrage. On the sunniest day (11 August, solar at 37% of load), prices fell to $-\text{€}60/\text{MWh}$ in the early afternoon. That trough came at the hours the typical day predicted, only much deeper. The value was large, and a schedule learned from the calendar would have found it.

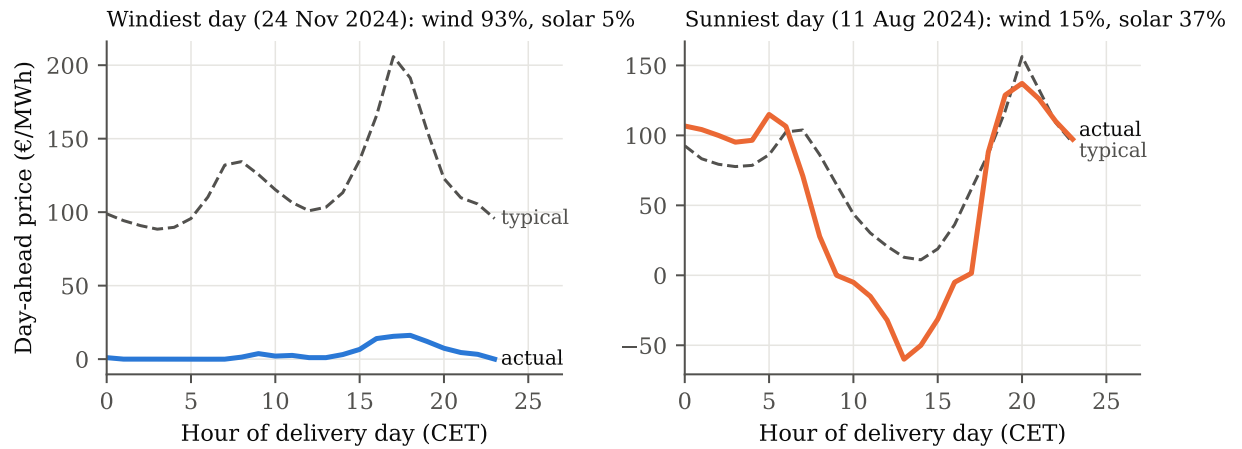


Figure 1: Two days in DE_LU, 2024: day-ahead prices (solid) against the typical-day profile of the previous 28 days (dashed). Left: the day with the highest forecast wind penetration. Right: the day with the highest forecast solar penetration.

The typical-day operator, which knows the recent calendar shape and nothing about the day, captures on average 73.4% of the ceiling on the days where a capture share is defined. Persistence captures 64.0%. An operator that trusts yesterday's prices does worse than one that averages the last four weeks, because a single day's idiosyncrasies are more often noise than signal. The forecaster, which adds a correction built from the TSOs' day-ahead fundamentals, captures 76.0%. Day-specific information is therefore worth 2.6 points of capture on average over the calendar alone.

That gain is unevenly distributed (Figure 2). It is largest in Germany-Luxembourg (+5.8 points), Spain (+7.1) and Italy North (+6.5). In all four Nordic and Baltic zones (Finland, both Swedish zones and Estonia) the forecaster does slightly worse than the calendar, by 0.4 to 1.2 points. Norway is an outlier on every measure, for the reason above.

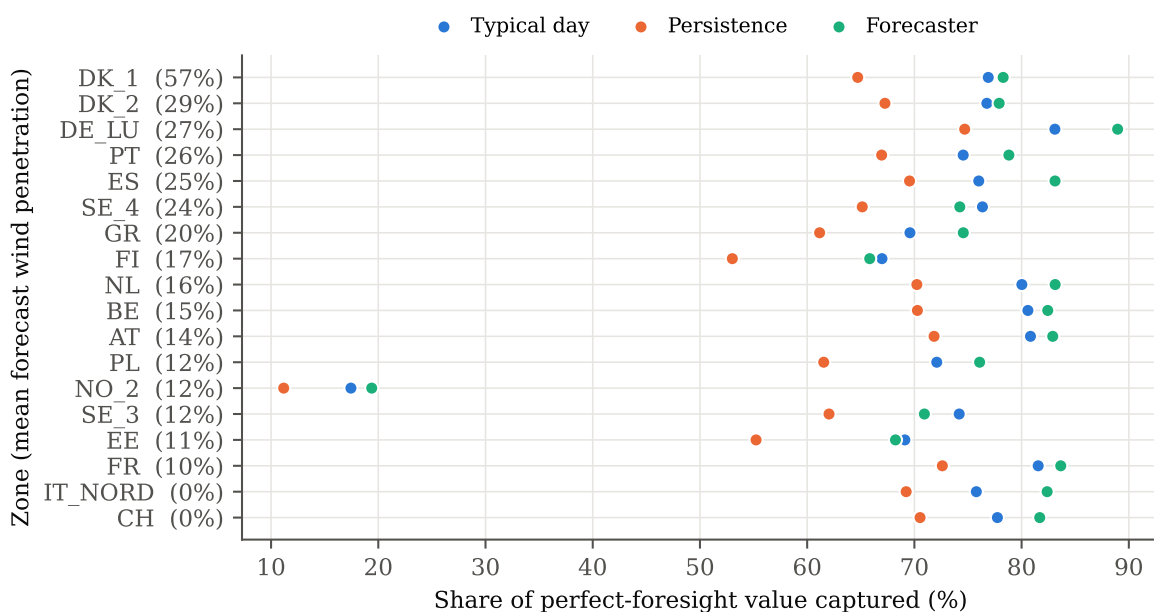


Figure 2: Share of the perfect-foresight value captured by each operator, by zone. Zones are ordered by mean forecast wind penetration, shown in brackets.

5.2 Which renewable creates arbitrage value

Table 2 reports the pre-registered specification for every outcome. Each column regresses the outcome on forecast wind and solar penetration, per 10 percentage points of forecast load, with zonexyear , zonexmonth and date fixed effects.

Table 2: The effect of forecast wind and solar penetration on storage outcomes.

	TD capture	PS capture	FC capture	log V*	Novelty	Spread (€/MWh)
Wind penetration (per 10 pp)	-0.21 (0.53)	-0.89 (0.72)	-0.98 (0.55)	-0.02 (0.02)	0.01*** (0.00)	-2.96*** (1.24)
WCB p-value	[0.718]	[0.368]	[0.203]	[0.248]	[0.000]	[0.003]

	TD capture	PS capture	FC capture	log V*	Novelty	Spread (€/MWh)
Solar penetration (per 10 pp)	4.15 (2.93)	5.53* (3.24)	3.70 (2.93)	0.25*** (0.05)	-0.05*** (0.01)	22.14*** (3.57)
WCB p-value	[0.157]	[0.092]	[0.203]	[0.001]	[0.001]	[0.000]
Wind – solar	-4.36* (2.59)	-6.42** (2.71)	-4.69* (2.58)	-0.27*** (0.06)	0.06*** (0.01)	-25.10*** (4.19)
WCB p-value	[0.099]	[0.029]	[0.068]	[0.000]	[0.000]	[0.000]
Mean of outcome	73.42	63.95	76.00	4.58	0.22	90.43
Zone-days	43,829	43,754	43,640	45,876	46,884	46,884
Zones	18	18	18	18	18	18

Notes: Each column is one regression of the outcome on forecast wind and solar penetration with zone×year, zone×month and date fixed effects. Capture shares are in percent of V*; novelty is in correlation units. The ‘Wind – solar’ row is the wind coefficient minus the solar coefficient, from the re-parameterisation in Section 4. Standard errors clustered by zone in parentheses; stars use wild cluster restricted bootstrap p-values (Webb weights, 9,999 draws), shown in brackets. * p<0.1, ** p<0.05, *** p<0.01.

The clearest result concerns the scale of the opportunity (columns 4 and 6, Figure 3). Within a zone’s year and season, and relative to other zones on the same day:

- **Solar creates arbitrage value.** 10 pp more forecast solar raises the day’s perfect-foresight value by 28% (log V* +0.249, WCB p = 0.001) and widens the daily spread by €22/MWh (p < 0.001).
- **Wind does not.** Its effect on value is small and insignificant (–2.0% per 10 pp, p = 0.25), and it *narrows* the spread by €3/MWh (p = 0.003).

The two technologies differ in how much they vary from day to day. After the fixed effects, the residual standard deviation of wind penetration is 11.8 pp, against 2.9 pp for solar. Scaled to one standard deviation, a sunny day raises arbitrage value by 7.4% and a windy day lowers it by 2.4%.

The mechanism is visible in Figure 1. A windy day lowers the whole price curve through the merit order more than it reshapes it. A sunny day deepens the midday trough against an evening peak that solar cannot reach.

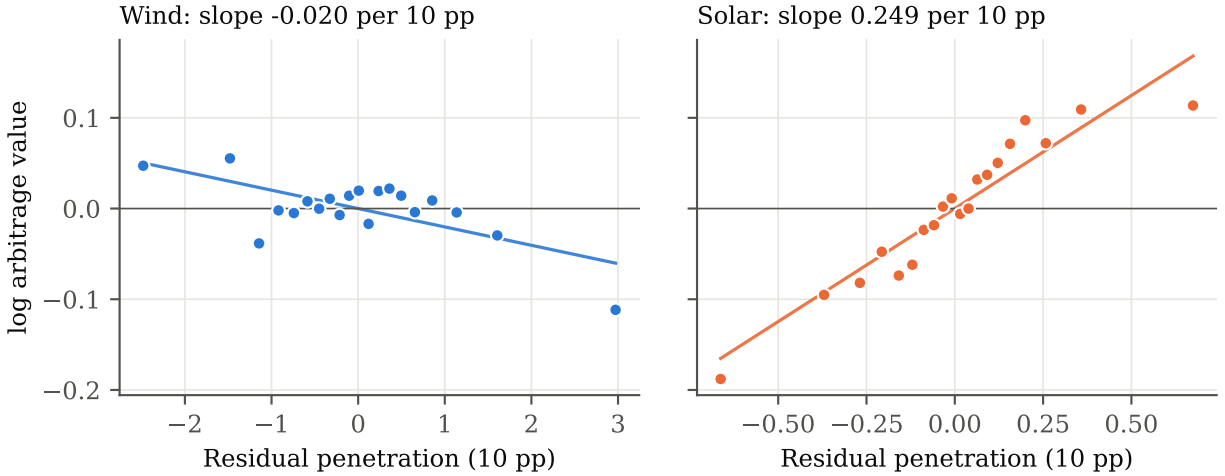


Figure 3: Binned scatter of log arbitrage value against residual forecast wind (left) and solar (right) penetration. Both variables are residualised on the other treatment and on $\text{zone} \times \text{year}$, $\text{zone} \times \text{month}$ and date fixed effects, so the slope equals the regression coefficient in Table 2. Each point is a vigintile of the residualised treatment.

5.3 The mechanism: how typical the day's shape is

The hypothesis concerns not how much value there is but where in the day it sits. Shape novelty measures directly how far a day's price shape departs from the typical day. It moves exactly as hypothesised (Table 2, column 5; Figure 4):

- **Wind makes the day less typical:** +0.013 per 10 pp, WCB $p < 0.001$.
- **Solar makes it more typical:** -0.051 per 10 pp, $p = 0.001$.
- **The difference** is 0.064 ($p < 0.001$).

Per standard deviation of weather the two effects are equal and opposite, at +0.016 and -0.015, each about 7% of the mean novelty of 0.22. Sunny days reinforce the shape the calendar predicts; windy days depart from it. This is the “sun is a clock” half of the hypothesis, measured on prices rather than on revenue.

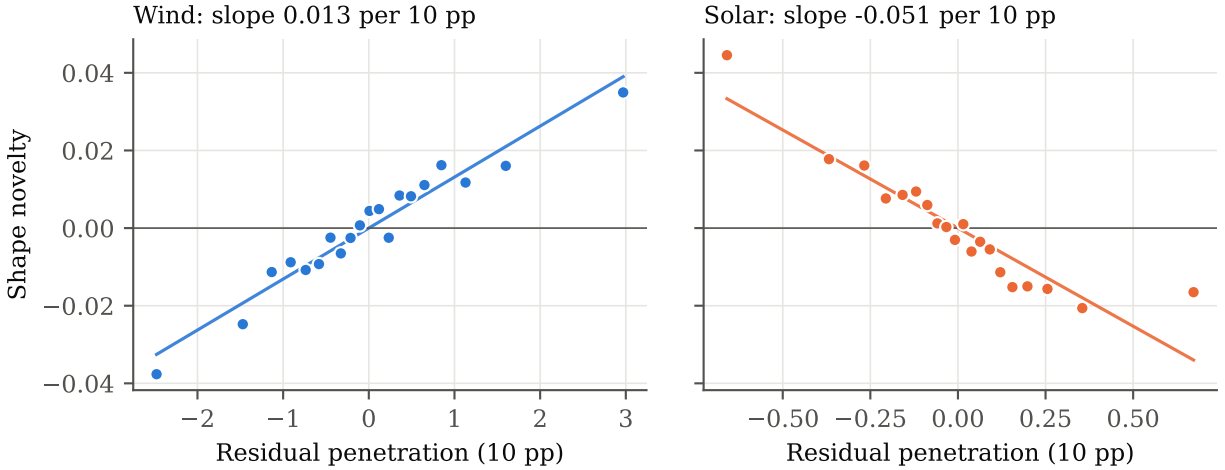


Figure 4: Binned scatter of shape novelty, $1 - \text{corr}(\text{day's prices, typical-day profile})$, against residual forecast wind (left) and solar (right) penetration. Construction as in Figure 3.

5.4 Capture: the pre-registered test

Our key hypothesis (H1) was that wind lowers the share of value capturable from the calendar more than solar does, so $\beta_w < \beta_s$ for typical-day capture. **The estimate has the predicted sign but is only marginally significant.** The difference is -4.36 points of capture per 10 pp (standard error 2.59, WCB $p = 0.099$, Table 2 column 1).

It also decomposes differently from the hypothesis as phrased. Wind barely moves typical-day capture (-0.21 , $p = 0.72$). The difference comes from solar *raising* it ($+4.15$, $p = 0.16$). The same pattern holds for the other operators:

Operator	$\beta_w - \beta_s$	WCB p
Persistence	-6.42	0.029
Forecaster	-4.69	0.068

Figure 5 shows the typical-day relationship.

We read this as consistent with the mechanism, though not conclusive about capture. Solar makes arbitrage value larger and more calendar-shaped, and calendar-based operators capture a larger share of it. Wind makes the day's shape less typical, but on average across the panel, it does not measurably reduce what a calendar operator captures. Section 5.5 shows that this average conceals heterogeneity by system. We did not register a one-sided test, so we report the two-sided p -value as registered.

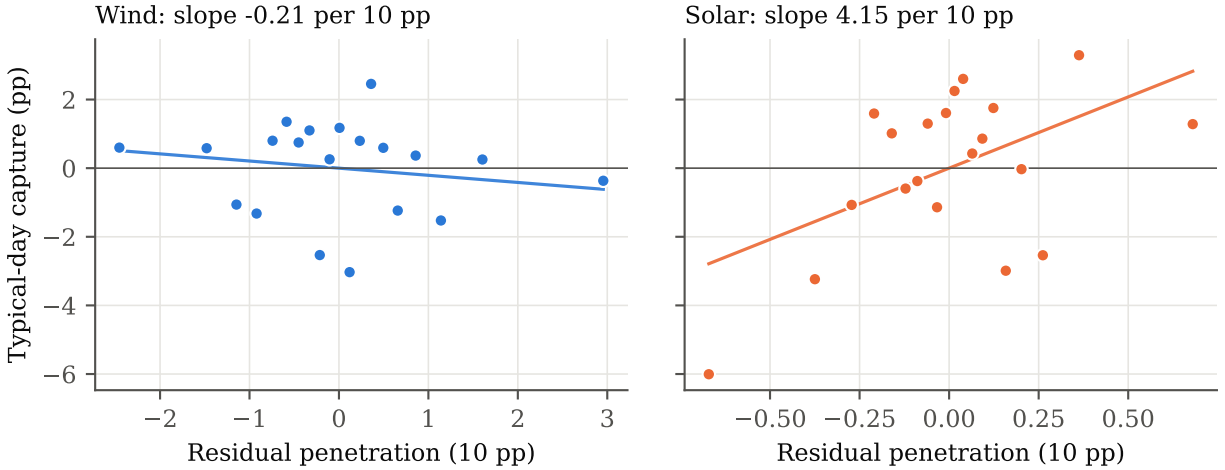


Figure 5: Binned scatter of typical-day capture share against residual forecast wind (left) and solar (right) penetration. Construction as in Figure 3.

Table 3 shows why the date fixed effects are essential, and how an unconditional analysis would have misled. With only zone and season effects, the solar coefficient is roughly twice as large. The wind-minus-solar difference is then -8.75 , and conventionally “significant at 1%.” Adding date effects halves both. Much of the naive association comes from days that are sunny across Europe at once, and those coincide with Europe-wide conditions, such as fuel prices and continental demand, that independently shape spreads. A single-market time-series analysis cannot remove that common component, which is why this question needs a panel.

Table 3: How the estimate moves as fixed effects are added.

	None	Zone	+ Zone×month	+ Zone×year	+ Date (main)
Wind penetration (per 10 pp)	-0.17 (0.51)	-0.59 (0.42)	-0.72** (0.31)	-0.49 (0.39)	-0.21 (0.53)
Solar penetration (per 10 pp)	8.32** (2.90)	7.04*** (0.66)	8.94*** (1.39)	8.25*** (1.49)	4.15 (2.93)
Wind – solar	-8.49*** (2.55)	-7.63*** (0.54)	-9.65*** (1.16)	-8.75*** (1.30)	-4.36 (2.59)
Mean of outcome	73.42	73.42	73.42	73.42	73.42
Zone-days	43,833	43,833	43,833	43,833	43,829
Zones	18	18	18	18	18

Notes: Outcome: typical-day capture share (%). Fixed effects added left to right. Standard errors clustered by zone; conventional p-values. The movement across columns shows which confounders each set of fixed effects removes.

5.5 Heterogeneity: wind's cost grows with wind's share

The pre-registered heterogeneity test (H4) interacts forecast wind penetration with the zone-year's mean wind penetration K . The interaction is -0.35 points of typical-day capture per 10 pp of wind, per additional 10 pp of K (standard error 0.08, WCB $p = 0.02$).

Evaluated at the panel mean ($K = 18.4\%$), a windy day has no significant effect on calendar capture ($+0.48$, s.e. 0.50). In a low-wind system with $K = 5\%$, it slightly raises it, by about $+0.9$ points per 10 pp. In a system with Denmark West's wind share ($K \approx 58\%$), it lowers it by about 0.9 points per 10 pp.

So the informational cost of wind is not a constant. It appears as wind comes to dominate a system, which is the structural direction the introduction asked about. One caution applies. K is not randomly assigned, so this is heterogeneity in the effect of weather, not the causal effect of building wind capacity. High-wind systems differ from others in interconnection, hydro access and market design, and any of these could modulate how wind weather reaches prices.

6 Robustness and limitations

6.1 The forecast-error placebo

The day-ahead auction clears before the forecast error is realised, so the error should not affect any day-ahead outcome. Table 4 adds realised minus forecast wind to the main specification.

- **The placebo passes for capture, value and spread.** The error coefficient is 0.05 for typical-day capture (WCB $p = 0.91$), -0.003 for $\log V^*$ ($p = 0.89$) and -2.5 for the spread ($p = 0.28$). This supports the timing of the data, and it supports the claim that these outcomes respond to what was known when the auction cleared.
- **It fails for shape novelty.** The error coefficient is 0.009 per 10 pp ($p = 0.014$), comparable in size to the forecast coefficient of 0.013.

Table 4: Placebo: the realised forecast error in wind.

Outcome	Forecast wind	Wind forecast error	WCB p (error)	Zone-days
TD capture	-0.210	0.046 (0.389)	[0.910]	43,779
$\log V^*$	-0.020	-0.003 (0.022)	[0.892]	45,825
Novelty	0.013	0.009 (0.003)	[0.014]	46,833
Spread	-2.940	-2.531 (2.326)	[0.282]	46,833

Notes: Adds realised minus forecast wind (per 10 pp of forecast load) to the main specification. The day-ahead auction clears before the error is realised, so its coefficient should be zero. Standard errors clustered by zone; WCB p -value for the error coefficient.

The failure has a natural interpretation, and it qualifies the wind-novelty result. Market participants trade on their own weather forecasts, not only the TSO's. When realised wind exceeds the

TSO forecast, commercial forecasts will often have anticipated part of the excess. So the error is partly a proxy for information the market had and our regressor lacks. On this reading, the TSO forecast measures the market’s information with error. The wind-novelty coefficient then reflects wind weather broadly, including the part the TSO failed to forecast, rather than the TSO forecast narrowly. The *direction* of the result, that windy weather makes a day’s shape less typical, is unaffected, since the forecast and its error act in the same direction. But its magnitude should not be read as the effect of the published forecast alone. The measurement-error interpretation also implies that the wind coefficients elsewhere in Table 2 are, if anything, attenuated.

6.2 Pre-registered robustness

Table 5 reports every robustness check fixed in the pre-registration, for the key outcome.

Table 5: Robustness of the key estimate (typical-day capture).

Specification	β wind	β solar	β wind – β solar	p (diff)	Zone-days
Baseline (2h)	-0.21 (0.53)	4.15 (2.93)	-4.36 (2.59)	0.099	43,829
1h battery	-0.16 (0.49)	4.23 (2.58)	-4.40 (2.34)	0.072	43,826
4h battery	-0.22 (0.80)	4.96 (4.24)	-5.18 (3.65)	0.147	43,824
Ratio floor 1st pct	-0.34 (0.99)	4.07 (4.60)	-4.41 (3.89)	0.261	45,464
Ratio floor 10th pct	-0.06 (0.33)	4.66 (2.19)	-4.73 (1.97)	0.029	41,700
Excluding gas crisis	-0.06 (0.33)	4.91 (1.78)	-4.97 (1.64)	0.008	31,434
Two-way cluster (zone, date)	-0.21 (0.52)	4.15 (2.85)	-4.36 (2.52)	0.102	43,829

Notes: Outcome: typical-day capture share (%), per 10 pp of forecast penetration. All specifications include zone×year, zone×month and date fixed effects. p-values are wild cluster restricted bootstrap (Webb, 9,999 draws) except the two-way-clustered row, which reports the conventional p-value.

The point estimate of $\beta_w - \beta_s$ is stable across every specification, from -4.36 to -5.18 points per 10 pp. It holds across battery durations of one, two and four hours, across ratio floors, and after excluding the 2021–23 gas crisis.

Precision is not stable, and the reason is informative. The estimate is sharpest where the ratio outcome is least noisy:

- Excluding the gas crisis gives $p = 0.008$.
- A 10th-percentile floor on the ceiling gives $p = 0.029$.
- A 1st-percentile floor, which admits more near-zero-value days whose capture ratios are erratic, gives $p = 0.26$.

Two-way clustering by zone and date barely changes the standard error (2.52 against 2.59). That suggests residual cross-sectional correlation on a given day is not driving inference.

The leave-one-zone-out estimates in Figure 6 range from -3.08 (without Poland) to -6.45 (without Norway). No single zone drives the sign. Norway’s reservoir-hydro zone contributes most of the imprecision. Without it, the estimate is -6.45 with a standard error of 1.44 . Norway has no solar, so it identifies only the wind coefficient, and its near-flat prices make its capture ratios the noisiest in the panel.

We do not promote any of these variants to the headline. The pre-registered estimate is the baseline, and it is marginal. What the robustness exercise shows is that its magnitude is not an artefact of any one choice, and that its imprecision comes from ratio noise on low-value days rather than from fragility in the sign.

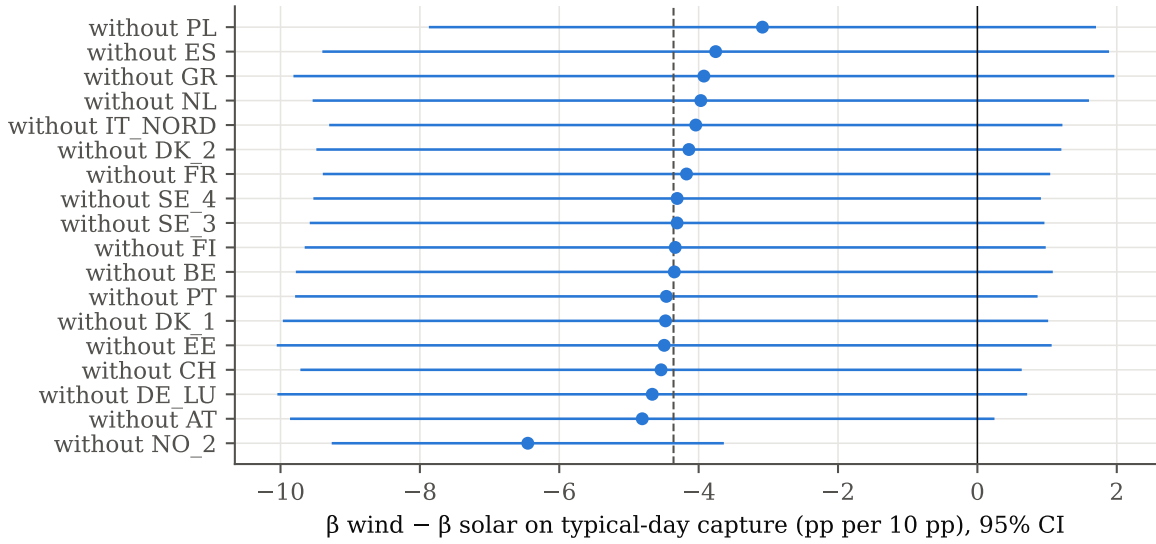


Figure 6: Leave-one-zone-out estimates of $\beta_w - \beta_s$ for typical-day capture, with 95% confidence intervals from zone-clustered standard errors. The dashed line is the full-sample estimate.

6.3 Limitations

Day-ahead only. Batteries in practice earn much of their revenue in intraday, balancing and ancillary-service markets, which this paper does not model. Our results concern the day-ahead arbitrage component. Intraday markets are where forecast errors are traded, so the informational content of wind may be larger there than here. That is the natural extension.

Price-taking and daily independence. The battery does not move prices, which is appropriate for one battery and not for a fleet (Butters, Dorsey, and Gowrisankaran 2025; Karaduman 2023). Each day starts and ends empty, which forgoes multi-day arbitrage. That understates value most for the four-hour battery and in hydro-dominated zones.

Forecast timing. Regulation requires TSO wind and solar forecasts by 18:00 on D-1, after the 12:00 auction. The forecaster’s information set is therefore mildly optimistic. That caveat touches only the forecaster operator. The typical-day and persistence operators use prices alone. The treatments are regressors rather than operator inputs, so this timing issue does not affect them.

Market coupling. Coupled zones’ prices respond to their neighbours’ weather (Lago et al. 2018). Date fixed effects remove the Europe-wide component of weather, but not regional spillovers. Our coefficients are best read as the effect of wind or solar weather in a zone’s region.

Few clusters. Eighteen zones is a moderate number of clusters. We use the wild cluster restricted bootstrap with Webb weights throughout (Webb 2023), but inference at this cluster count remains less reliable than with many.

One forecaster. The forecaster is one reasonable gradient-boosted model, not the frontier of price forecasting (Lago et al. 2021). Its capture is a lower bound on what day-specific information is worth.

Heterogeneity is descriptive. The finding that wind’s cost grows with wind’s share (Section 5.5) compares systems that differ in many ways besides wind capacity. It does not identify the causal effect of adding capacity.

7 Conclusion

This paper asked whether the two technologies of decarbonisation create storage value of different kinds: value available to anyone who knows the calendar, or value available only to those who forecast the day. Across 19 European bidding zones, with identification from weather under zone-by-year, zone-by-month and date fixed effects, we find three things.

1. **Solar creates arbitrage value, and wind does not.** A day with 10 percentage points more forecast solar offers 28% more perfect-foresight value and a €22/MWh wider spread. A windier day offers no more value and a slightly narrower spread, because wind lowers the whole price curve more than it reshapes it.
2. **Solar makes the day’s price shape more typical, and wind makes it less typical.** Per standard deviation of weather, the two effects are equal and opposite. This is the mechanism the hypothesis proposed, and it is the paper’s most precisely estimated result.
3. **Calendar-based operators capture a larger share of solar-driven value.** The difference between wind and solar has the predicted sign and a stable magnitude of about four to five points of capture per 10 pp. At the pre-registered specification it is only marginally significant ($p = 0.10$). It is stronger for persistence operators and in wind-dominated systems, where wind’s informational cost emerges most clearly.

The sun, in short, behaves like a clock. Whether the wind behaves like a lottery depends on how much of it a system has.

Implications.

- **Solar-dominated systems.** The storage value that decarbonisation creates in systems like Iberia, Greece and northern Italy is largely *public* in our sense: it recurs at predictable hours, so simple operators can capture most of it, and forecasting capability confers little advantage. Entry into storage there should face low informational barriers.

- **Wind-dominated systems.** In systems like Denmark, windy days erode what calendar rules capture, and the returns to forecasting rise with wind's share. There, the market structure of flexibility is more likely to favour operators with scale in data and trading. TSO forecasts, already published as a public good, are then worth more to small operators.

Extensions. Three follow directly from this design:

- **Intraday and balancing markets,** where wind's forecast errors are traded, and where its informational content is likely to be larger.
- **An equilibrium treatment of storage entry,** since a fleet large enough to flatten the solar trough would change the shape it trades on.
- **The Iberian gas cap of 2022–23,** a genuine natural experiment in price formation within the same data.

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A Pre-registration and deviations

All hypotheses, the specification, the sample, the outcome definitions and the robustness set were fixed in a pre-registration document committed to the project’s version control before any panel backtest had run (commit 6aa7cdc, 27 September 2026). The registered hypotheses were:

	Outcome	Prediction
H1	Typical-day capture	$\beta_w < \beta_s$
H2	Shape novelty	$\beta_w > 0$ and $\beta_w > \beta_s$
H3	$\log V^*$	No directional prediction
H4	Typical-day capture	The wind effect is more negative where wind is already a larger share
Placebo	All of the above	The forecast-error coefficient is zero

Five deviations were recorded in the pre-registration’s log. Each was recorded before the first regression was estimated.

1. **Czechia leaves the regression sample.** It publishes no day-ahead wind forecast, so wind penetration is undefined. Imputing zero would invent data.
2. **The gas-crisis window is fixed at 1 July 2021 to 30 June 2023.** The registration named the robustness check but not its dates.
3. **The forecaster predicts deviations from the typical-day profile, not price levels.** It therefore nests the typical-day operator, and tree ensembles cannot extrapolate levels outside their training range.
4. **Capture shares require $V^* > 0$.** NO_2 has 609 days with perfectly flat prices, on which a share is undefined rather than zero. The 5th-percentile floor is computed among positive days. This changes the floor only for NO_2.
5. **Implausible load forecasts are set to missing.** These are days where the forecast is outside 50–200% of the zone’s median for that month. The rule flags 11 days, all reporting errors.

Outcomes against predictions.

- **H1:** correct sign, marginal ($p = 0.099$).
- **H2:** holds, $p < 0.001$.
- **H3:** solar raises V^* and wind does not.
- **H4:** holds, $p = 0.02$.
- **Placebo:** passes for three of four outcomes and fails for novelty, as discussed in Section 6.1.

Why the design was built this way. An earlier, single-market analysis of the same question on British data had reported that decarbonisation shifts storage value towards sophisticated operators. That claim did not survive the addition of year fixed effects. Renewable penetration and forecaster

capture both trended upward over the sample, and the specification could not separate them. The current design rules that failure out in three ways:

- identification comes from weather within zone-years, so no common trend can load onto the treatment;
- the treatments are forecasts, which is the information on which the auction clears;
- the specification was fixed before the data were examined.

B Operator details

The optimisation. All operators solve the same mixed-integer programme, with the HiGHS solver through SciPy, on the price vector their information set provides. The schedule is then settled at actual prices. The panel required approximately 640,000 solves ($19 \text{ zones} \times 3 \text{ durations} \times 4 \text{ operators} \times \text{about } 2,800 \text{ days}$) with no solver failures. Each schedule was checked against the physical limits on state of charge and power before being recorded.

The forecaster. For each zone, a histogram gradient-boosted regression tree predicts, for every hour of day d , the deviation of the price from the typical-day profile. Its features are:

- the market hour, weekday and month;
- the typical-day price for that hour;
- yesterday's price at that hour and yesterday's mean, each as a deviation from the profile;
- the TSO's day-ahead forecasts of load, wind and solar for that hour, and each relative to its own 28-day mean;
- the day's forecast wind and solar shares of load.

The model is refitted monthly on an expanding window of all earlier days, and it forecasts only months strictly after its training data. The first forecasts use models trained on 2018. Germany-Luxembourg's zone begins only in October 2018, and the first model needs 180 days of training, so its forecasts begin on 1 May 2019.

Reproducibility. Every table, figure and quoted number is produced by code in the project repository, from the raw ENTSO-E data, and the paper is compiled from those outputs:

```
python -m gbmo.ingest.entsoe && python -m gbmo.ingest.load_zones
python -m gbmo.arbitrage.panel && python -m gbmo.arbitrage.panel_forecast
python -m gbmo.analysis.panel_data && python -m gbmo.analysis.results
python paper/build.py
```